Action-Conditioned Predictive Diagnostics for JEPA World Models

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Abstract

Visual changes that leave the underlying control state unchanged can still alter the futures predicted by a world model. Encoder-level invariance alone does not reveal whether such discrepancies are amplified by learned dynamics or matter to downstream planning. We study this problem with Action-Conditioned Predictive Consistency (ACPC), which compares nominal and visually perturbed histories after rolling them forward under the same action sequence; the score itself uses neither future targets nor behavior labels. ACPC yields exact diagnostic inequalities for future-error and squared goal-cost drift, together with a different-state margin guard against trivial collapse. Across four control tasks, eight-step ACPC reduces mean absolute error for predicting future-error drift by $51.3 \pm 3.5\%$ relative to the strongest matched action-destroyed control. Candidate-conditioned ACPC is also informative about planner cost drift and decision regret. Source-task calibration transfers unchanged to the remaining tasks with 0.84–0.86 balanced accuracy, while one fixed rule retains useful checkpoint ordering under blur and resize without stressor-specific adjustment. The same analysis is instantiated on PLDM with model-family-specific numerical calibration. Together, these results support action-conditioned rollout disagreement as a target-free signal of predictive sensitivity under visual shifts.

Keywords: visual world models; predictive stability; action-conditioned diagnostics; calibration transfer; visual corruption.

Code and data release: `https://github.com/Anguo-star/lewm-acpc-diagnostics`

1 Introduction

World models support control by predicting the outcomes of candidate actions [11, 12]. Joint-embedding predictive architectures (JEPAs) replace pixel reconstruction with prediction in representation space [16, 1, 3]. This can reduce pressure to model task-irrelevant visual detail [28, 17], which is attractive when texture, lighting, or image quality changes without changing the control state.

Robust control, however, does not call for indiscriminate invariance. Background texture may be a nuisance, but a contact pose, doorway state, or object–goal relation must remain distinguishable when it changes the useful action. A planner also consumes predictions rather than an encoder output alone: a small encoding discrepancy can grow through learned dynamics, while a large one can contract before the cost readout. The relevant question is therefore whether the discrepancy introduced by a same-state visual perturbation remains small after the same action sequence.

Consistency must also be selective. Mapping every observation to the same point would make nominal and perturbed inputs agree perfectly while destroying information needed for action choice. A useful diagnostic must combine same-state stability with evidence that genuinely different states remain separated [8, 33, 24, 10]. To inform planning, it should also indicate when predictive disagreement can change candidate costs or decisions.

We study these requirements with Action-Conditioned Predictive Consistency (ACPC). This is a diagnostic study of frozen checkpoints, not a new robust-training method. ACPC rolls a nominal history and a visually perturbed version of the same state through the predictor under a shared action sequence, measuring projected latent disagreement without a realized future or behavior label. For each pair, exact inequalities bound changes in future prediction error and, for squared-distance goals, candidate cost. A different-state margin guard rejects trivial low-radius collapse. The same predictive interface can be used across inspectable models, although numerical thresholds may remain model-family specific.

We evaluate ACPC on LeWM [19] under Gaussian observation noise across TwoRoom, PushT, Reacher, and Cube, and instantiate the same analysis on a complete PLDM sweep as a second architecture. No-noise LeWM checkpoints remain strong on clean observations yet lose up to 74.56 ± 5.55 success points at evaluation noise $\sigma = 0.08$. We first test whether eight-step recorded-action ACPC explains future-error drift beyond one-step and same-horizon zeroed/shuffled-action controls. We then test candidate-conditioned measurements against planner cost drift and decision regret. Finally, we study calibration over all source/remaining-task splits, apply one LeWM rule to blur and resize without stressor-specific adjustment, and run the same analysis on PLDM without assuming shared numerical thresholds.

The contributions are:

1. **A target-free diagnostic of predictive sensitivity.** We define ACPC by rolling nominal and visually perturbed histories forward under the same actions. Exact bounds relate its disagreement score to future-error and goal-cost changes, while a different-state margin rules out trivial collapse.
2. **Controlled evidence for prediction and planning relevance.** Across four tasks and training runs, eight-step ACPC predicts future-error drift better than one-step and matched zeroed/shuffled-action controls. Candidate-conditioned ACPC is also informative about planner cost drift and decision regret.
3. **Transfer across tasks, stressors, and architectures.** We evaluate every source/remaining-task partition, apply one LeWM calibration to blur and resize without stressor-specific adjustment, and instantiate the full analysis on PLDM. These tests separate portability of the analysis from portability of numerical thresholds, which remain model-family specific.

2 Related Work

Latent-predictive world models. JEPA [16], I-JEPA [1], and V-JEPA [3, 2] learn by predicting latent targets rather than reconstructing pixels. LeWM [19], our primary family, stabilizes an end-to-end JEPA world model with SIGReg [7]; PLDM [25, 26], implemented through `stable-worldmodel` [18], supplies a distinct latent-dynamics architecture. Theory explains why latent prediction can suppress nuisance or noisy features [28, 17, 14], but does not make a learned checkpoint robust to every action-relevant visual shift. Seq-JEPA addresses the related invariance–equivariance tension through model design [9]; ACPC instead audits how a frozen predictor propagates paired visual differences under actions.

Visual robustness and control-relevant representations. Data augmentation methods such as DrQ/DrQ-v2 [15, 32] and SODA [13] improve visual control by changing training. ViGMO uses latent consistency for generalization to unseen visual distractions, while VIBR learns view-invariant value functions for robust visual control [20, 6]. ReOI instead intervenes on observations at test time to obtain distractor-robust world-model predictions for visual MPC [5]. ACPC uses paired probes to diagnose a frozen checkpoint; it neither retrains the model nor repairs the input. Its selective-consistency motivation is related to bisimulation, bisimulation-based MPC, and value-aware models, which preserve distinctions with different downstream control consequences [8, 33, 24, 10, 29, 27]. Our narrower intervention asks whether two views of the same state remain close after one shared action sequence while a proxy guard rejects gross representational collapse.

Action and dynamics consistency diagnostics. Recent work uses inverse probes, latent action decoding, cycle consistency, or generated-future compatibility to test action semantics [4, 34, 23, 21, 30]. MWM incorporates action-conditioned consistency during training [31], whereas iKCE diagnoses kinematic long-horizon failures [22]. ACPC instead intervenes on paired visual views of the same state and connects rollout disagreement to action-matched future-error drift; we do not claim priority for action consistency in general.

3 Target-Free Predictive Diagnostic

3.1 Action-matched rollout radius

Let h be a nominal observation–action history and $\tilde{h}$ a visually probed view of the same underlying state. A frozen encoder E_θ maps the histories to $z = E_\theta(h)$ and $\tilde{z} = E_\theta(\tilde{h})$. Both branches are propagated by the predictor F_θ under the *same* action sequence $\mathbf{a} = (a_0, \ldots, a_{H-1})$. For a task-relevant projection Π and nonnegative weights

$\sum_{k=1}^{H} \alpha_k = 1$, define

$$\bar{G}_{\mathbf{a}}(z) = \left[\sqrt{\alpha_1}\Pi(F_\theta^1(z, a_0));\ \cdots;\ \sqrt{\alpha_H}\Pi(F_\theta^H(z, \mathbf{a}))\right], \qquad \mathrm{ACPC}_H(h, \tilde{h}, \mathbf{a}) = \|\bar{G}_{\mathbf{a}}(z) - \bar{G}_{\mathbf{a}}(\tilde{z})\|_2. \tag{1}$$

The encoder distance is identical for every candidate action sequence; ACPC can additionally expose action-dependent amplification of that displacement. Whether this extra information matters is an empirical question, tested against one-step and action/time-destroyed controls rather than assumed from the definition.

Theorem 1 (Target-free prediction-error-drift bound)**.** *Let $Y_{\mathbf{a}}^H$ be any projected external future generated from the same state and action sequence as both prediction branches. With*

$$e_h = \|\bar{G}_{\mathbf{a}}(E_\theta(h)) - Y_{\mathbf{a}}^H\|_2, \qquad e_{\tilde{h}} = \|\bar{G}_{\mathbf{a}}(E_\theta(\tilde{h})) - Y_{\mathbf{a}}^H\|_2,$$

the following holds sample by sample:

$$|e_{\tilde{h}} - e_h| \le \mathrm{ACPC}_H(h, \tilde{h}, \mathbf{a}). \tag{2}$$

The adverse increase $(e_{\tilde{h}} - e_h)_+$ obeys the same bound, while the right-hand side requires no future target at diagnostic time.

Proof. This is the reverse triangle inequality in the weighted-stacked projected-rollout space. The common action sequence is required because both errors must share the same target. □

For planning, the relevant object is the final predicted latent for each candidate. Write $x_j = \Pi(F_\theta^H(E_\theta(h), \mathbf{a}^j))$ and $\tilde{x}_j = \Pi(F_\theta^H(E_\theta(\tilde{h}), \mathbf{a}^j))$. Its displacement $r_j = \|x_j - \tilde{x}_j\|_2$ is the final-step ACPC component; whenever $\alpha_H > 0$, $r_j \le \mathrm{ACPC}_H/\sqrt{\alpha_H}$.

Theorem 2 (Goal-cost drift, fixed-pool regret, and certificates)**.** *LeWM scores candidate j against a fixed goal embedding g by $C_j = \|x_j - g\|_2^2$ and $\tilde{C}_j = \|\tilde{x}_j - g\|_2^2$. Define*

$$b_j = r_j\big(\|x_j - g\|_2 + \|\tilde{x}_j - g\|_2\big). \tag{3}$$

Then $|\tilde{C}_j - C_j| \le b_j$ for every candidate. If w and $\tilde{w}$ are the nominal and perturbed winners of the same pool, respectively, then the clean-history decision regret satisfies

$$0 \le C_{\tilde{w}} - C_w \le b_{\tilde{w}} + b_w. \tag{4}$$

If w is unique and

$$\min_{j \ne w}\big(C_j - C_w - b_j - b_w\big) > 0, \tag{5}$$

the perturbed branch has the same winner. More generally, if $\mathcal{E}$ is the nominal top-k elite set and

$$\min_{i \in \mathcal{E},\, j \notin \mathcal{E}}\big(C_j - C_i - b_j - b_i\big) > 0, \tag{6}$$

the perturbed branch has exactly the same elite set.

Unlike a global Lipschitz argument, Equation (3) is computed from the actual paired candidate endpoints and adapts to each candidate's distance from the goal. It is worst-case tight from these quantities alone; proofs and equality cases are in Appendix A.

Corollary 1 (Conditional adaptive-CEM alignment)**.** *Consider nominal and perturbed CEM runs initialized by the same proposal and sampled with common random numbers. While their ordered candidate pools are aligned, if Equation* (6) *holds at the current iteration, both runs fit the same next proposal. If it holds at every iteration, the pools, proposals, and final action remain aligned by induction.*

The corollary is deliberately one-sided: after an elite certificate fails, later pools need not be comparable and ACPC supplies no adaptive guarantee.

3.2 Radius, discriminability guard, and calibration

For anchor i, let s_i be the median norm of its clean observed transitions, including the history-to-future boundary:

$$s_i = Q_{0.50}\big(\{\|u^{\mathrm{obs}}_{i,k} - u^{\mathrm{obs}}_{i,k-1}\|_2\}_{k=1}^{H}\big)\,. \tag{7}$$

For probe draw m, the normalized radius and checkpoint-level *Action-Conditioned Trajectory Radius* (ATR) statistic are

$$R^{(i,m)} = \frac{\|\bar{G}_{\mathbf{a}_i}(E_\theta(h_i)) - \bar{G}_{\mathbf{a}_i}(E_\theta(\tilde{h}_i^{(m)}))\|_2}{s_i + \varepsilon}, \qquad \mathrm{ATR}_q = Q_q\left(\left\{\frac{1}{M}\sum_{m=1}^{M} R^{(i,m)}\right\}_{i=1}^{n}\right). \tag{8}$$

We use $H = 8$, uniform horizon weights, and $q = 0.90$. The unnormalized numerator is the target-free quantity used by the theory and planner panel. For cross-task checkpoint screening, ATR reuses each fixed logged anchor's clean transition scale; this scale is not an outcome label, but it makes normalized ATR an offline logged-data audit rather than an online single-history alarm. ATR is an empirical upper-tail summary, not a candidate-distribution probability.

Low radius alone accepts a constant model, so consistency must be selective. Let $\mathcal{P}_{\mathrm{same}}$ contain nominal/probed histories of the same state and $\mathcal{P}_{\mathrm{diff}}$ contain task-defined proxy pairs known to cross a state distinction. Denote their normalized rollout distances by d_θ.

Definition 1 (Selective discriminability)**.** For radius r, margin $\gamma > 0$, and pass probabilities $p_{\mathrm{same}}, p_{\mathrm{diff}}$, a representation is selectively discriminative when

$$\Pr_{\mathcal{P}_{\mathrm{same}}}[d_\theta \le r] \ge p_{\mathrm{same}}, \qquad \Pr_{\mathcal{P}_{\mathrm{diff}}}[d_\theta \ge r + \gamma] \ge p_{\mathrm{diff}}. \tag{9}$$

A constant representation can satisfy the first condition but violates the second for every positive margin. Selective Margin Pass Rate (SMPR) is the empirical fraction of different-state proxy pairs satisfying the second inequality when r is the same-state q90 tube. It is a gross-collapse guard, not a proof that every task-relevant distinction is preserved.

The construction requires paired views, an inspectable encoder–predictor path, and one common projection and normalization within a comparison. It does not assume LeWM-specific layers, losses, or representation geometry. For model family $\mathcal{F}$ and stressor υ, calibration thresholds t_R, t_M define the higher-is-better score

$$S_{\mathcal{F},\upsilon}(\theta) = \min\left\{\frac{t_R - \mathrm{ATR}_\upsilon(\theta)}{|t_R| + \varepsilon}, \frac{\mathrm{SMPR}_\upsilon(\theta) - t_M}{|t_M| + \varepsilon}\right\}. \tag{10}$$

The endpoint rule $S \ge 0$ uses only one checkpoint and no behavior label at scoring time on the fixed anchor audit, but its thresholds are family-calibrated. Given a reference checkpoint θ_0, the paired rule

$$\Delta S_{\mathcal{F},\upsilon}(\theta_1;\theta_0) = S_{\mathcal{F},\upsilon}(\theta_1) - S_{\mathcal{F},\upsilon}(\theta_0) > 0 \tag{11}$$

asks whether the endpoint improves under the same scoring map. Its zero threshold survives a common monotone rescaling within a pair, but the rule requires the reference and does not imply that either checkpoint is absolutely robust.

Definition 2 (Family-calibrated diagnostic region)**.** The diagnostic region is the set of checkpoints satisfying $\mathrm{ATR} \le t_R$ and $\mathrm{SMPR} \ge t_M$ under one fixed family/stressor calibration. Its meaning is established empirically on tasks excluded from threshold selection; raw thresholds are not assumed to transfer across model architectures.

Four summaries used below should not be conflated. *Encoder-shift q90* measures the 90th percentile latent change before any rollout and is only an upstream comparator. *One-step ACPC q90* rolls both views forward by one predicted step. *Eight-step action-matched ACPC q90* rolls both views autoregressively for eight predicted steps under the same recorded actions; its normalized checkpoint summary is ATR. The *selective ACPC score* in Equation (10) combines ATR with the SMPR guard. Here H always denotes the number of predicted future steps, not the number of history frames, and *action matched* means that the two branches receive the same recorded or candidate action sequence. ACPC/ATR and the selective radius–margin construction are our diagnostic objects; encoder shift and the one-step quantity are comparison features.

Appendix G gives the frozen fixed-pool and adaptive-CEM evaluation contract, including the point at which the conditional induction ceases to apply.

4 Experimental Setup

Models, tasks, and stressors. LeWM [19] is an end-to-end JEPA world model used for latent-space model-predictive control and is the primary family for the mechanism, planner, and cross-stressor panels. PLDM [25, 26] supplies a second inspectable latent-dynamics architecture for the architecture-portability test. TwoRoom, PushT, Reacher, and Cube span navigation, planar manipulation, arm control, and 3D manipulation. For each architecture, the complete Gaussian sweep contains a no-noise checkpoint and full-sequence Gaussian-augmentation checkpoints with $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$. The primary endpoint adds observation noise $\sigma = 0.08$ while keeping the goal image clean. Off-axis tests use fixed Gaussian blur ($k = 15$) and resize (scale 0.25).

Behavior and diagnostic units. For each task and LeWM training seed 3072/3073/3074, the nine-point Gaussian grid is evaluated with seeds 42/43/44 and 100 trajectories per evaluation seed. Evaluation seeds are conditional measurement replicates; training seed is the model-replication unit. ATR uses 100 anchors, five probe draws, $H = 8$, uniform horizon weights, within-anchor draw averaging, and the checkpoint q90. SMPR uses the q90 tube, closest-35% state-proxy neighborhood, and margin 0.10. Protocol details are in Appendix B. The PLDM architecture check uses one independently trained model family over the same four tasks and nine training-noise levels, with the same evaluation seeds 42/43/44 and 100 trajectories per seed. This complete 36-checkpoint grid supplies the architecture instantiation and cross-task analysis; evaluation replicates are not counted as additional trained models.

Future-drift prediction. The mechanism test uses 16 trajectory blocks per task, four probe severities, and two probe draws. Every regression predicts the same target: the absolute change in eight-step latent prediction error between the nominal and probed histories. A block is predicted by a ridge model fitted on the other 15 blocks, and this rotation is repeated for all blocks. Common features are probe severity, encoder shift, action magnitude, and nominal dynamics. The first comparison adds one-step action-matched ACPC; the second adds the strongest eight-step control with zeroed, cross-trajectory, or time-shuffled actions; the proposed model instead adds eight-step ACPC computed with the recorded action sequence. Thus the target horizon is fixed—the experiment tests information content, not the unsurprising accumulation of rollout error with horizon. All four tasks and training seeds 3072/3073/3074 use the same protocol.

Planner-aligned panel. On seeds 3072/3073/3074 base and noise-trained endpoint checkpoints, the fixed-pool panel evaluates 100 independent histories per task, role, and training seed with the actual ordered step-0 CEM pool ($K = 64$, top eight). The adaptive panel starts both branches from the same proposal and common random numbers for eight CEM steps; a full-budget panel uses the deployed $K = 300$, top-30, 30-step setting on the same 16 fixed histories per task, reused across roles and training seeds. Within each training seed and for each evaluation task, ridge models are fitted on the other three tasks. We ask whether candidate-conditioned five-step ACPC, matching the candidate evaluation horizon, improves prediction of cost drift, first-action RMS, or clean-history decision regret beyond severity, checkpoint role, candidate-conditioned one-step ACPC, and nominal top-1 margin. This is a finite-budget mechanism audit over three independent training runs, not an estimate of a broad training-run population.

Cross-task threshold selection. We enumerate every nonempty proper subset of the four tasks as a threshold source: four one-task sources, six two-task sources, and four three-task sources. For each of the 14 directional partitions, (t_R, t_M) is selected using all nine Gaussian checkpoints and all three training seeds from the source tasks, then applied unchanged to every remaining task. Metrics first average the three training runs within each evaluation task and then give each task equal weight; checkpoint rows are never treated as independent replicates. Behavioral recovery labels are used only to select and evaluate the rule, never as score inputs.

Cross-stressor transfer. For each evaluation task, we use the threshold selected from the other three Gaussian tasks. We then score all 24 fixed-severity base–endpoint pairs (four tasks, three training seeds, blur and resize) without stressor-specific adjustment. Positive behavioral change requires at least five percentage points of stressed-success gain and no more than five points of clean-success loss. The analysis compares the change in the *final selective score* with behavioral change. It deliberately does not rank encoder shift, one-step ACPC, or other components under the new stressors; their comparison belongs to the Gaussian mechanism experiments, not to this transfer question.

5 Experiments

Gaussian evaluation noise first creates a controlled fragile/recovered family. We then test action-matched future drift, planner-facing cost and decision stability, threshold transfer across task subsets, and transfer to blur/resize in that order. Closed-loop success supplies behavioral labels but is never an input to the diagnostic score.

5.1 Behavioral recovery and diagnostic co-movement

At evaluation noise $\sigma = 0.08$, no-noise LeWM checkpoints lose 25.89 ± 2.38, 74.56 ± 5.55, 41.00 ± 1.44, and 22.11 ± 2.78 success points on TwoRoom, PushT, Reacher, and Cube, respectively. Full-sequence Gaussian augmentation recovers broad, task-dependent regions rather than one universal best training strength. Figure 1 shows the observation-noise score together with ATR and SMPR over the complete nine-level training sweep.

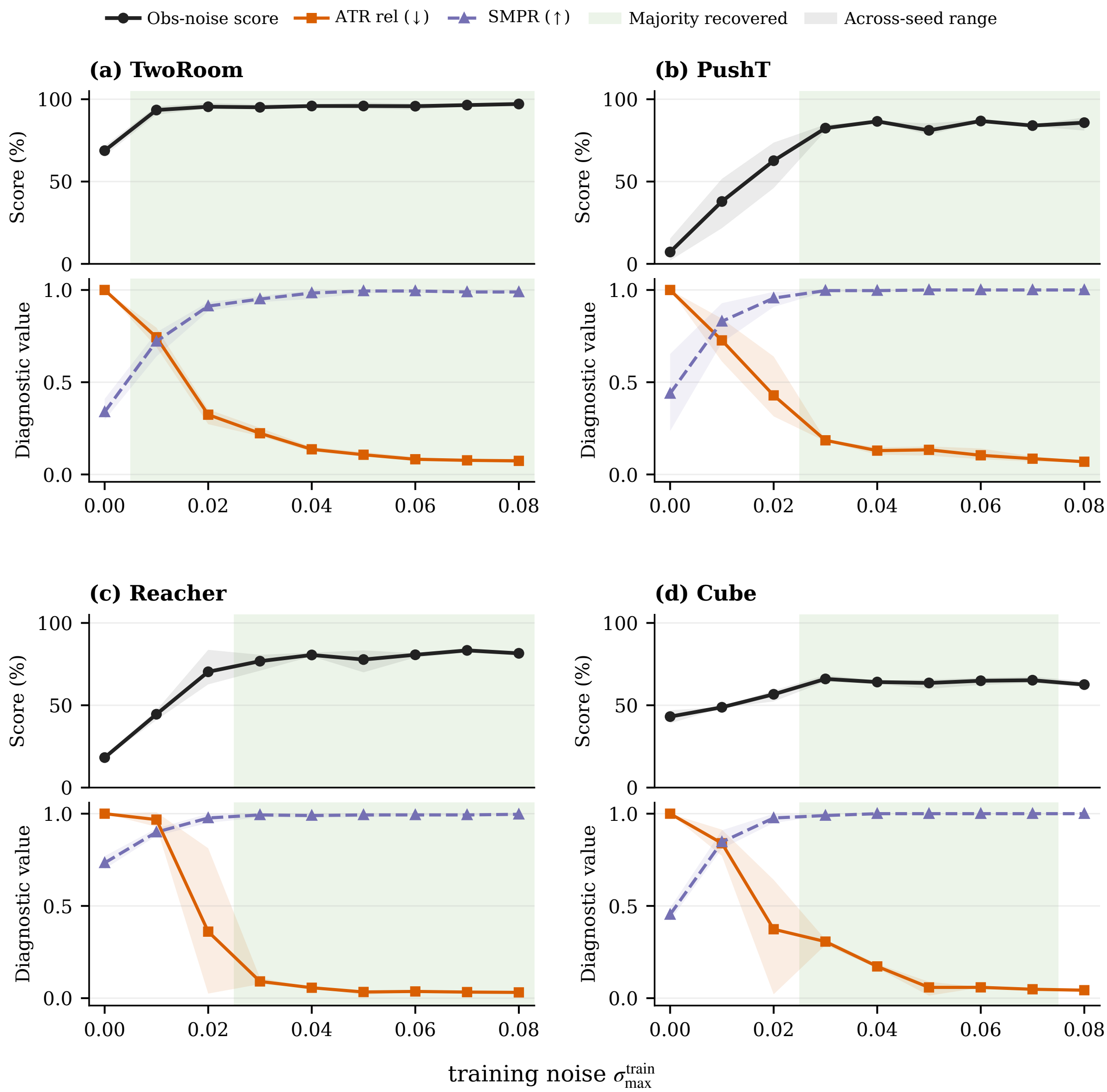

Figure 1: Closed-loop success under observation noise and the two diagnostic components across the complete Gaussian training sweep. Solid lines first average evaluation replicates within each training run and then average runs; translucent envelopes span the run means. ATR is divided by the corresponding task×run no-noise value for display (base = 1); SMPR remains on its original rate scale. Green bands require at least 80% of the attainable stressed-score gain, at most a five-point clean loss, and recovery in a majority of runs. Lower ATR and higher SMPR co-move with recovery on all four tasks, but this descriptive alignment alone does not establish action or semantic specificity.

ATR contracts and SMPR rises through the recovery region on every task. The panel motivates a calibrated two-component screen, but descriptive co-movement alone does not establish action specificity and behavior remains the outcome authority.

5.2 Action-matched rollouts predict future drift

Theorem 1 applies to action-matched rollouts, but a useful mechanism test must separate action information from rollout length. Every regression here predicts the same eight-step latent future-error-drift target. The feature sets differ: common covariates plus one-step action-matched ACPC; common covariates plus the strongest eight-step zeroed/shuffled-action control; or common covariates plus eight-step ACPC computed with the recorded actions.

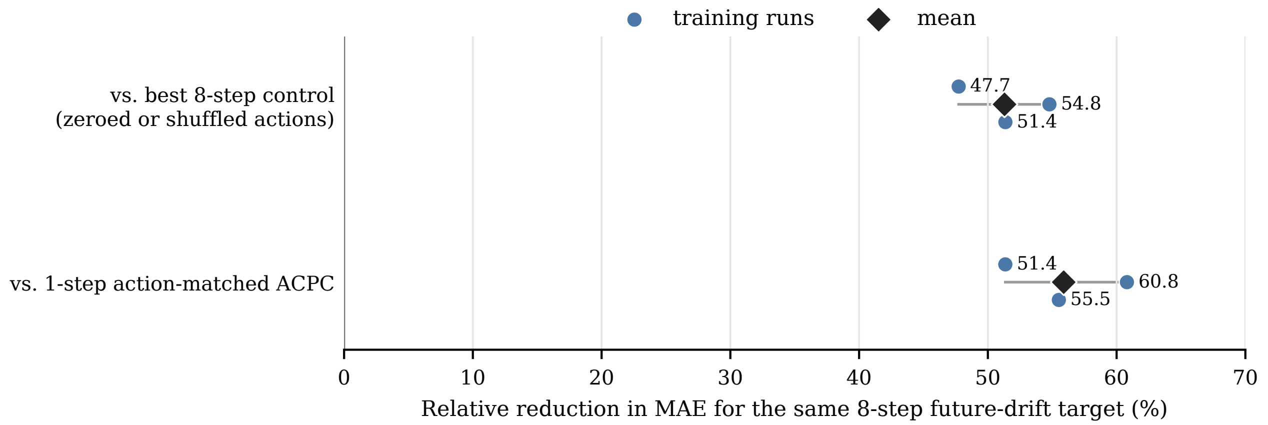

Figure 2: Recorded-action eight-step ACPC adds information about a common eight-step future-drift target. Points are equal-task MAE reductions for independently trained models; diamonds are their means. The upper comparison holds rollout horizon fixed and changes only action structure, so the result cannot be explained by longer rollouts mechanically accumulating more error. Every model improves on all four tasks.

Relative to one-step ACPC, recorded-action eight-step ACPC reduces equal-task MAE by $55.9 \pm 4.7\%$ across training runs. Relative to the strongest same-horizon action-destroyed control, the reduction is $51.3 \pm 3.5\%$. Every task–run comparison improves. These percentages are reductions in error for predicting the fixed future-drift target; they are not changes in closed-loop success and not a claim that an eight-step prediction is intrinsically more accurate than a one-step prediction. Appendix C gives exact task-level MAEs and control identities.

5.3 ACPC connects to planner cost and decision stability

The deterministic result gives a shared-pool regret ceiling, but its sufficient certificates may abstain and adaptive pools can diverge. We therefore measure both fixed-pool cost drift and adaptive decisions. Across 9,600 joined reduced-budget history records, the squared-distance bound has zero violations, and neither the top-1 nor elite-set certificate passes falsely. Identity probes produce zero five-step ACPC and zero first-action RMS; whenever the adaptive induction is active, all proposal updates remain aligned.

Table 1: Cross-task ridge analysis on 9,600 matched reduced-budget history records. Models are fitted separately within each training run on three tasks and evaluated on the fourth. The reference features are severity, checkpoint role, candidate-conditioned one-step ACPC, and nominal top-1 margin; the expanded model adds candidate-conditioned five-step ACPC. Entries are equal-task MAE reductions; mean ± sample SD treats the training run as the model-replication unit, and the final column counts improving task–run evaluations.

Response	Mean ± SD	Run range	Cells improved
Fixed-pool max cost drift	7.9 ± 1.4%	6.6–9.5%	8/12
Adaptive positive decision regret	15.2 ± 2.0%	12.9–16.9%	12/12
Adaptive first-action RMS	1.1 ± 0.5%	0.8–1.7%	10/12

Adding candidate-conditioned five-step ACPC reduces fixed-pool maximum-cost-drift MAE by $7.9 \pm 1.4\%$ across training runs, although the gain remains task-dependent. Adaptive positive decision-regret MAE falls by $15.2 \pm 2.0\%$ and improves for every evaluated task and run. First-action-RMS MAE falls by only $1.1 \pm 0.5\%$, for which we make no material incremental claim. Rank association with the five-step quantity exceeds that of the one-step quantity for every evaluated task and run across all three responses.

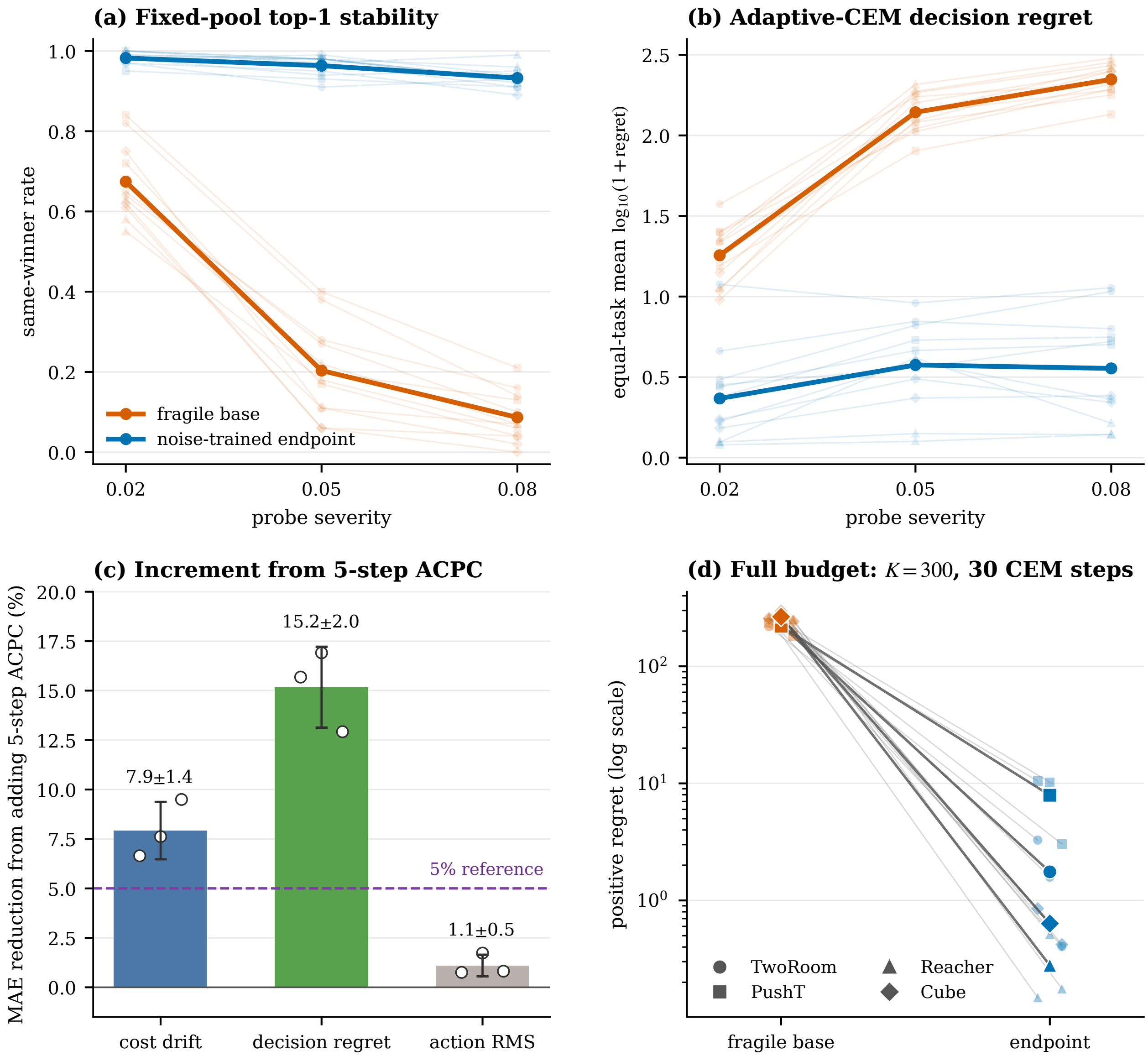

Figure 3: Planner-aligned ACPC evidence. Faint curves/points in (a), (b), and (d) show task–run blocks; heavy curves or points show equal-task–run means. Panels (a)–(c) use 100 histories per task, run, and checkpoint role: (a) same-winner frequency for the shared fixed pool, (b) adaptive-CEM regret of the perturbed decision when rescored on the clean history, and (c) run-level prediction-error reductions from adding candidate-conditioned five-step ACPC beyond severity, checkpoint role, one-step ACPC, and nominal margin; bars and error bars are mean ± sample SD across runs. Panel (d) reports the same regret outcome on the same 16 fixed histories per task, evaluated for every run and role at the deployed $K = 300$, 30-step budget. These are behavior-blind planner diagnostics, not closed-loop return.

At severity 0.08, task-level training-run mean fixed-pool top-1 stability is 0.92–0.96 for noise-trained endpoints versus 0.02–0.13 for fragile bases. Adaptive positive regret is lower for every endpoint task–run cell: its equal-task mean is 3.64 ± 0.68 versus 227.04 ± 16.92 at reduced budget and 2.65 ± 1.43 versus 244.89 ± 27.08 at full budget (mean ± sample SD across runs). Absolute cost scales differ by task, so these base–endpoint contrasts are descriptive rather than a pooled causal effect. Per-task values and numerical details appear in Appendix G.

5.4 Thresholds transfer across task subsets

We test all 14 directional source partitions rather than selecting one favorable split (Fig. 4).

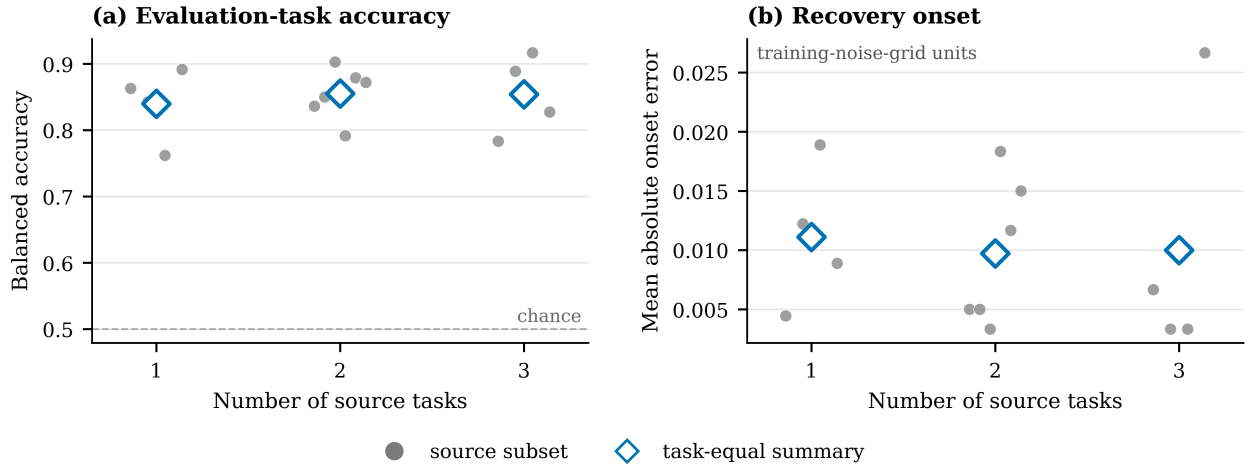

Figure 4: Cross-task portability over every source/evaluation partition. For each source subset, ATR and SMPR thresholds are selected using only those tasks and then applied unchanged to all remaining tasks. Small points are individual source subsets; large points give task-equal summaries. Adding source tasks is not monotonically beneficial because composition matters, but the aggregate rule remains stable across one-, two-, and three-task sources.

Balanced accuracy is 0.840, 0.855, and 0.854 when thresholds are selected from one, two, and three source tasks, respectively. Corresponding precision/recall values are 0.903/0.824, 0.904/0.865, and 0.904/0.862. Mean absolute recovery-onset errors are 0.011, 0.010, and 0.010 training-noise-grid units, with maximum error 0.030 in each source coverage level. Performance does not improve monotonically with the number of source tasks: source composition is consequential, especially when TwoRoom is the evaluation task. This is evidence for portability of the selective diagnostic principle, not for a universal numerical threshold. Appendix D reports every partition and retains every training run in each task.

5.5 PLDM separates analysis portability from numerical calibration

We next run the same paired-history construction, eight-step rollout, normalization, SMPR guard, recovery definition, and threshold-selection algorithm on the complete PLDM Gaussian sweep. No PLDM behavior label is used as a score input, and no LeWM-specific layer or training loss enters the analysis. For each PLDM evaluation task, numerical thresholds are selected using the other three PLDM tasks and then applied unchanged to its nine checkpoints.

Pooling the four evaluation branches gives 0.836 balanced accuracy, 0.789 precision, and 0.882 recall. Task balanced accuracies are 0.938, 0.750, 0.500, and 0.875 for TwoRoom, PushT, Reacher, and Cube, respectively, so the Reacher boundary case is retained rather than averaged away. After PLDM-specific task-relative anchoring, reusing the corresponding LeWM source-split thresholds gives 0.807 balanced accuracy, 0.778 precision, and 0.824 recall. This is a score-aligned portability reference, not evidence for a universal raw threshold. Together, the analyses support an architecture-portable mathematical and analysis interface with family-specific numerical calibration.

Table 2: PLDM architecture-portability check on the complete four-task, nine-checkpoint Gaussian sweep. Both rows use the current task-relative ATR score: thresholds are selected on the other three PLDM tasks or transferred from the corresponding LeWM split. Metrics pool all 36 evaluation rows. Raw numerical thresholds are not assumed to be shared across model families.

Calibration	BA	Precision	Recall	False pass	False miss
PLDM-local, other three tasks	0.836	0.789	0.882	4	2
LeWM-source reference, PLDM-anchored	0.807	0.778	0.824	4	3

5.6 The selective score retains ordering under blur and resize

We apply the other-three-task Gaussian threshold to every task's blur/resize pairs without stressor-specific selection. This tests the final ATR+SMPR score, not its individual components.

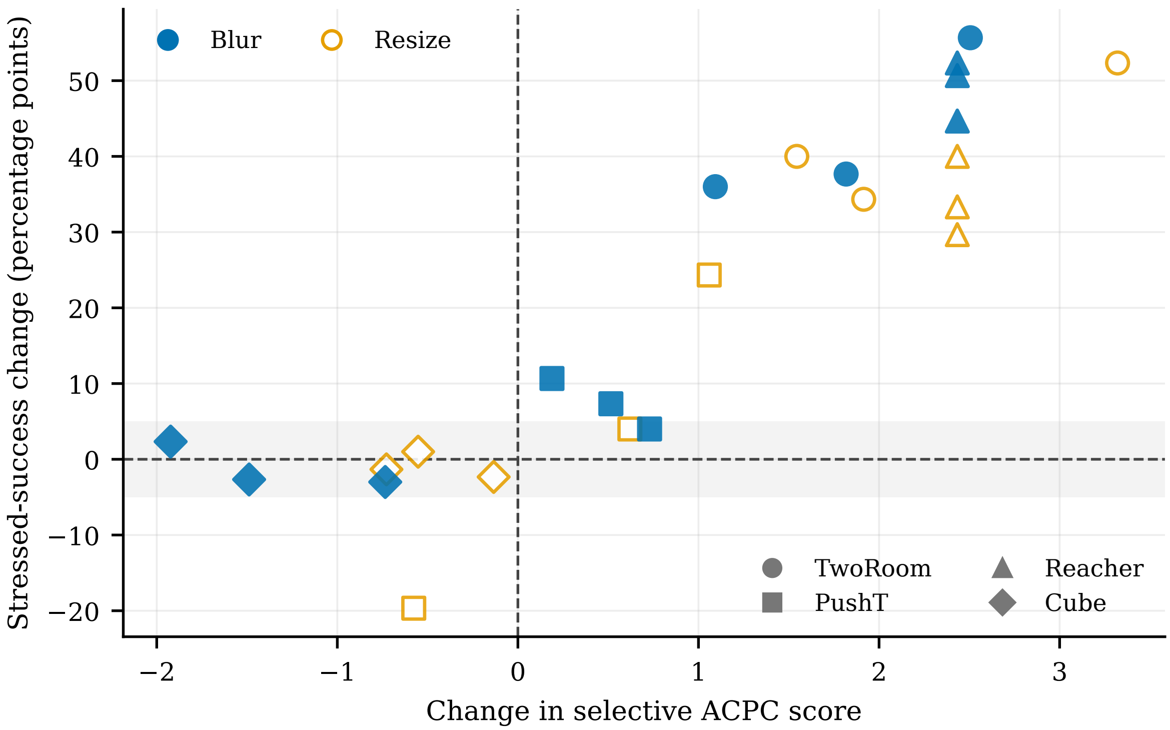

Figure 5: Cross-stressor ordering for all 24 LeWM pairs. The horizontal axis is the endpoint-minus-base change in the selective ACPC score using thresholds from the other three Gaussian tasks; the vertical axis is the corresponding blur/resize success change. Marker shape denotes task and fill denotes stressor. No blur/resize row is used to select a threshold.

The sign rule reaches 0.889 balanced accuracy, 0.882 precision, and 1.000 recall, while the continuous score change has 0.909 Spearman correlation with stressed-success change. Blur and resize separately reach 0.875/0.900 balanced accuracy. The two discordant cases are both PushT changes of exactly four percentage points, which fall below the fixed five-point positive label while the selective score improves. These results support relative ordering across these two stressors at one severity; they do not turn the score into an absolute cross-stressor robustness certificate.

6 Discussion and limitations

The future-drift experiment validates the object in Theorem 1: eight-step ACPC computed with the recorded actions contains information about action-matched latent error drift that an upstream shift, one-step ACPC, and same-horizon action-destroyed features miss. Because all regressions predict the same eight-step target, this result is not the trivial statement that rollout errors grow with horizon. The direction is consistent across all four tasks and independent training runs, which supports repeatability of the conclusion but does not provide a precise estimate of broad training-run population variance.

The planner panel closes a different link. The candidate-cost inequality is deterministic, and all certificate invariants hold on the evaluated pools; candidate-conditioned five-step ACPC adds information for adaptive clean-history regret beyond the one-step quantity and nominal margin consistently across tasks and training runs. The cost-drift increment is positive on average but more task-dependent, while the increment is negligible for first-action RMS. Independent training runs support repeatability of this finite-budget mechanism, not precise training-run population inference or a closed-loop robustness gain. The deployed-budget check reuses 16 fixed history blocks per task across models, and the audit does not include matched nominal-only timing, so it establishes neither a precise history-population estimate nor incremental wall-clock overhead.

Cross-task calibration gives the strongest generalization result: every one-, two-, and three-task source subset is evaluated on all remaining tasks. The absence of monotonic improvement with source-set size cautions against a single universal threshold and shows that task composition matters. The blur/resize analysis asks a narrower question—whether the final selective score preserves relative checkpoint ordering—and does not re-use it as evidence for action specificity.

The PLDM experiment addresses architecture scope directly. The ACPC inequalities, rollout statistic, collapse guard, and cross-task calibration procedure execute unchanged on a second latent-dynamics architecture. PLDM-local calibration and task-relative LeWM transfer answer different questions. The first tests the analysis method on PLDM; the second tests a score that already contains task-relative anchoring. Their contrast supports architecture portability with family-specific calibration, not a LeWM-only method or a universal raw threshold.

Across the experiments, SMPR remains a gross-collapse guard rather than a semantic proof, the cost certificates require candidate-wise paired evaluation, and the adaptive result is conditional on pool alignment until an elite certificate fails. Stable-but-wrong predictions remain possible. Closed-loop evaluation is therefore the behavioral authority. Important next steps are closed-loop replanning outcomes, multiple severities for each off-axis stressor, richer different-state proxies, and additional model families with different objectives and planner interfaces.

7 Conclusion

ACPC converts a paired visual probe into testable bounds on action-matched future-error and goal-cost drift, while a different-state margin guard rules out trivial collapse. Across four tasks, its recorded-action long-horizon response adds information beyond one-step and same-horizon action-destroyed features. The resulting selective rule transfers across every task-subset split and retains checkpoint ordering under blur and resize without stressor-specific adjustment. The same analysis is instantiated on PLDM, while its numerical thresholds remain architecture-local. ACPC is therefore an architecture-portable, family-calibrated predictive diagnostic with explicit operational boundaries, not a universal robustness score.

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A Candidate-Cost and Fixed-Pool Analysis

Proof of Theorem 2. For one candidate, suppress the index and factor the squared-distance difference:

$$\begin{aligned}
|\tilde{C} - C| &= \big| \|\tilde{x} - g\|_2^2 - \|x - g\|_2^2 \big| && (12)\\
&= \big| \langle \tilde{x} - x, \tilde{x} + x - 2g \rangle \big| && (13)\\
&\le \|\tilde{x} - x\|_2 \, \|\tilde{x} + x - 2g\|_2 && (14)\\
&\le r\big(\|x - g\|_2 + \|\tilde{x} - g\|_2\big) = b. && (15)
\end{aligned}$$

The first inequality is Cauchy–Schwarz and the second is the triangle inequality. Thus, for nominal winner w and competitor j,

$$\tilde{C}_j - \tilde{C}_w \ge C_j - C_w - |\tilde{C}_j - C_j| - |\tilde{C}_w - C_w| \ge C_j - C_w - b_j - b_w.$$

For the perturbed winner $\tilde{w}$,

$$C_{\tilde{w}} \le \tilde{C}_{\tilde{w}} + b_{\tilde{w}} \le \tilde{C}_w + b_{\tilde{w}} \le C_w + b_w + b_{\tilde{w}},$$

which proves Equation (4); nonnegativity follows from the optimality of w on the nominal pool. Positivity of Equation (5) makes every perturbed competitor strictly more costly than w. The same argument for every $i \in \mathcal{E}$ and $j \notin \mathcal{E}$ proves that Equation (6) preserves the exact elite set. Strict inequalities avoid relying on implementation-specific tie breaking.

Proof of Corollary 1. At iteration zero, the proposals are identical by assumption. Common random numbers therefore produce the same ordered action pool. If the elite certificate holds, both branches select the same action vectors as elites. The CEM mean/variance update is a deterministic function of those vectors, so the next proposals are identical. Repeating this argument establishes pool and proposal alignment by induction. If a certificate fails, equality of the elite sets is unknown; the induction stops and no later shared-pool statement is made.

Relation to the weighted rollout radius. The planner bound uses the displacement at the cost readout horizon. For the weighted stack in Equation (1), $\sqrt{\alpha_H}\, r_j \leq \mathrm{ACPC}_H$ whenever $\alpha_H > 0$. The formal planner panel records r_j directly, avoiding the looser $1/\sqrt{\alpha_H}$ conversion. This does not require a future target, but it does require evaluating the candidate under both matched histories.

Sharpness without additional assumptions. Neither principal bound admits a uniformly smaller right-hand side from the available quantities alone. For a unit vector u, common target $Y = 0$, and two stacked predictions au and bu with $a, b \geq 0$, the reverse-triangle bound in Equation (2) holds with equality. Likewise, setting $g = 0$, $x = au$, and $\tilde{x} = bu$ makes $|\tilde{C} - C| = |b - a|(a + b) = r(\|x - g\|_2 + \|\tilde{x} - g\|_2)$.

B Evaluation and Diagnostic Protocol

These details define the fixed evaluation and diagnostic protocol used by the main tables.

LeWM is evaluated on PushT, TwoRoom, Reacher, and Cube using the canonical three evaluation seeds 42/43/44 and 100 trajectories per seed. The main Gaussian robustness endpoint perturbs observation pixels with Gaussian noise at standard deviation 0.08 while keeping the goal image clean. Noise-trained checkpoints use full-sequence input-side Gaussian augmentation with $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$.

PLDM uses one independently trained model family and the same four-task, nine-checkpoint Gaussian grid, evaluation seeds, trajectory count, paired history construction, $H = 8$ rollout statistic, and SMPR definition. For its architecture-local cross-task analysis, ATR is divided by the corresponding PLDM task's no-noise value and thresholds are selected on the other three PLDM tasks. The evaluation task's labels never enter threshold selection. A score-aligned transfer test applies the corresponding current LeWM threshold pair to each task-relative PLDM score. Raw numerical thresholds are not assumed to be shared across model families.

Within each task and training seed, let B and M be the stressed scores of the no-noise checkpoint and the best checkpoint in the frozen sweep. A row is "recovered" exactly when its stressed score is at least $B + 0.8(M - B)$ and its clean score is no more than five percentage points below the no-noise clean score. The green spans in Figure 1 require this label for a majority of the three training seeds.

ATR is computed at each task–training-seed–checkpoint row from one canonical weighted-stacked horizon radius per anchor. The default is $H = 8$ with uniform $\alpha_k = 1/H$. Each radius is divided by that anchor's q50 clean observed-transition scale, including the transition from the last history observation to the first future observation; multiple noise draws are first averaged within the anchor, and q90 is then taken across anchors. For the sweep display and family-local cross-task calibration, this checkpoint statistic is additionally divided by the corresponding task×seed no-noise value before thresholds or run summaries are computed. SMPR uses programmatic label-crossing state pairs, q90 tube radius, closest-35% neighborhood, and normalized margin 0.10. Across training seeds 3072/3073/3074, reported means and dispersions are descriptive rather than population inference.

For cross-task evaluation, candidate thresholds are $t_R \in \{0.05, 0.075, 0.10, 0.15, 0.20, 0.30\}$ for task-relative ATR and $t_M \in \{0.80, 0.85, 0.90, 0.95\}$ for SMPR. Within each source-task subset, the selection criterion lexicographically minimizes mean absolute onset error, false-early and false-late counts, then maximizes balanced accuracy. The selected pair is applied without modification to every evaluation task. No evaluation-task label or blur/resize outcome enters threshold selection.

All summary figures and tables are rebuilt deterministically by the scripts documented in `paper1/scripts/README.md`. Checkpoint-level diagnostics additionally require the released checkpoints and datasets. Machine-readable summaries record training seeds, evaluation seeds, and protocol hashes.

Table 3: Compact view of the complete nine-level Gaussian training sweep. "Best" selects the level with highest mean success under observation noise $\sigma = 0.08$; the full trajectories over all nine levels remain in Fig. 1. ATR is divided by each task×run no-noise value (base = 1) and is lower-is-better; SMPR remains on its original rate scale and is higher-is-better.

Task	Base success (%)	Best success (%)	Best $\sigma_{\max}$	ATR: base → best	SMPR: base → best	Majority-recovered levels
TwoRoom	68.8	97.1	0.08	1.00→0.07	0.34→0.99	0.01–0.08
PushT	7.2	86.8	0.06	1.00→0.10	0.44→1.00	0.03–0.08
Reacher	18.2	83.3	0.07	1.00→0.03	0.73→0.99	0.03–0.08
Cube	43.1	66.0	0.03	1.00→0.31	0.45→0.99	0.03–0.07

Table 4: Three independent training runs give the same future-drift conclusion. Every model predicts the same eight-step latent error-drift target. The first comparison adds eight-step ACPC computed with the recorded action sequence to common covariates plus one-step ACPC. The second replaces the one-step comparator by the strongest same-horizon control whose actions are zeroed or shuffled. Values are equal-task relative MAE reductions after rotating evaluation over 16 trajectory groups (fit on 15, evaluate the remaining group); higher is better.

Training run	vs. one-step ACPC	vs. same-horizon control	Task–run cells improved
seed 3072	55.5%	51.4%	4/4
seed 3073	60.8%	54.8%	4/4
seed 3074	51.4%	47.7%	4/4
mean ± SD	55.9 ± 4.7%	51.3 ± 3.5%	12/12

C Future-Drift Scope and Controls

The target-free inequality in Theorem 1 has zero numerical violations in every logged eight-step and candidate-conditioned five-step task×seed×checkpoint-role cell. This is an implementation invariant, not an independent statistical success count.

Logged-future prediction and candidate planning are different estimands: the former predicts an independent action-matched observed future after common covariates are supplied, whereas the latter compares candidates from the actual planner pool. We do not infer one result from the other. The separate panel in Section 5.3 supplies planner-facing evidence with same-pool, candidate-conditioned one- and five-step features.

The future-drift claim is restricted to no-noise base checkpoints under the listed visual probes. Seeds 3072, 3073, and 3074 are three independent model training runs and are reported symmetrically. Each passes all four tasks; the across-seed mean and sample standard deviation summarize repeatability rather than a precisely estimated population distribution.

D Cross-Task Threshold Partitions

The 14 rows below are exhaustive: selecting one, two, or three of four tasks as sources creates $4 + 6 + 4$ directional partitions. Each evaluation task retains all three training seeds and all nine checkpoints. The apparent sample size is therefore not inflated by treating checkpoint rows as independent observations.

E Cross-Stressor Pairs and Discordant Cases

Both discordant rows are near the behavioral labeling boundary: PushT seed 3072 under resize and PushT seed 3074 under blur each improve stressed success by four percentage points, one point below the fixed positive-transfer criterion, while their selective scores increase. The complete table makes these cases visible rather than reporting only aggregate accuracy.

Table 5: Absolute MAE for the common eight-step latent error-drift target. Each row contains 16 trajectory blocks. The recorded-action column is the proposed eight-step ACPC feature; the control column reports the strongest zeroed- or shuffled-action feature at the same horizon. Lower is better.

Task	seed	one-step	same-horizon control	recorded-action 8-step	control type	win blocks
TwoRoom	3072	2.535	2.099	**0.755**	time-shuffled actions	15/16
TwoRoom	3073	2.336	2.015	**0.866**	time-shuffled actions	15/16
TwoRoom	3074	2.075	1.911	**1.006**	cross-trajectory actions	13/16
PushT	3072	2.068	2.068	**1.762**	time-shuffled actions	11/16
PushT	3073	1.969	2.008	**1.315**	zero actions	13/16
PushT	3074	1.909	1.833	**1.439**	zero actions	13/16
Reacher	3072	1.358	1.097	**0.288**	time-shuffled actions	16/16
Reacher	3073	1.399	0.793	**0.247**	time-shuffled actions	16/16
Reacher	3074	1.409	1.136	**0.263**	time-shuffled actions	16/16
Cube	3072	1.171	1.037	**0.489**	zero actions	14/16
Cube	3073	1.416	1.212	**0.500**	zero actions	14/16
Cube	3074	1.061	1.008	**0.552**	time-shuffled actions	14/16

Table 6: Cross-task portability of the selective ACPC rule. Thresholds are selected only on the source tasks and applied unchanged to the evaluation tasks. Metrics first average three training runs within each task. Onset errors use training-noise-grid units. This table evaluates threshold portability, not individual diagnostic components.

Source tasks	Evaluation tasks	τ_R	τ_M	BA	P / R	Onset error mean / max
TwoRoom	PushT, Reacher, Cube	0.3	0.90	0.863	0.872 / 0.931	0.004 / 0.010
PushT	TwoRoom, Reacher, Cube	0.2	0.95	0.842	0.906 / 0.816	0.012 / 0.030
Reacher	TwoRoom, PushT, Cube	0.15	0.95	0.762	0.883 / 0.655	0.019 / 0.030
Cube	TwoRoom, PushT, Reacher	0.3	0.95	0.892	0.950 / 0.894	0.009 / 0.020
TwoRoom, PushT	Reacher, Cube	0.3	0.90	0.836	0.836 / 0.925	0.005 / 0.010
TwoRoom, Reacher	PushT, Cube	0.3	0.90	0.850	0.856 / 0.897	0.005 / 0.010
TwoRoom, Cube	PushT, Reacher	0.3	0.90	0.903	0.925 / 0.972	0.003 / 0.010
PushT, Reacher	TwoRoom, Cube	0.2	0.95	0.791	0.883 / 0.725	0.018 / 0.030
PushT, Cube	TwoRoom, Reacher	0.3	0.95	0.879	0.952 / 0.869	0.012 / 0.020
Reacher, Cube	TwoRoom, PushT	0.2	0.95	0.872	0.972 / 0.800	0.015 / 0.030
TwoRoom, PushT, Reacher	Cube	0.3	0.90	0.783	0.767 / 0.850	0.007 / 0.010
TwoRoom, PushT, Cube	Reacher	0.3	0.90	0.889	0.905 / 1.000	0.003 / 0.010
TwoRoom, Reacher, Cube	PushT	0.3	0.90	0.917	0.944 / 0.944	0.003 / 0.010
PushT, Reacher, Cube	TwoRoom	0.2	0.95	0.827	1.000 / 0.655	0.027 / 0.030

Table 7: Transfer of the selective ACPC score to blur and resize. For each task, thresholds are selected on the other three Gaussian tasks and applied without stressor-specific adjustment. The score change is compared with endpoint-minus-base stressed-success change; this evaluates relative ordering rather than absolute robustness detection.

Stressor subset	Pairs	BA	Precision / recall	Spearman	Discordant
All pairs	24	0.889	0.882 / 1.000	0.909	2
Blur	12	0.875	0.889 / 1.000	0.937	1
Resize	12	0.900	0.875 / 1.000	0.859	1

Table 8: All 24 LeWM blur/resize pairs used in the cross-stressor analysis. $\mathrm{ATR}_{\mathrm{rel}}$ and SMPR are endpoint values; ΔB is endpoint-minus-base stressed success in percentage points, and ΔS is the corresponding selective-score change. A behavioral change is positive only when $\Delta B \geq 5$ points and clean success loses at most five points; a score change is positive when $\Delta S > 0$. Outcomes are shown as behavior / score.

Task	Seed	Stressor	$\mathrm{ATR}_{\mathrm{rel}}$	SMPR	ΔB	ΔS	Outcome
TwoRoom	3072	blur	0.499	0.885	55.7	2.505	positive / positive
TwoRoom	3072	resize	0.336	0.967	52.3	3.321	positive / positive
TwoRoom	3073	blur	0.781	0.262	36.0	1.093	positive / positive
TwoRoom	3073	resize	0.691	0.361	40.0	1.544	positive / positive
TwoRoom	3074	blur	0.636	0.541	37.7	1.818	positive / positive
TwoRoom	3074	resize	0.617	0.656	34.3	1.915	positive / positive
PushT	3072	blur	0.943	0.939	10.7	0.189	positive / positive
PushT	3072	resize	0.814	0.969	4.0	0.620	nonpositive / positive
PushT	3073	blur	0.846	0.918	7.3	0.515	positive / positive
PushT	3073	resize	0.682	0.969	24.3	1.060	positive / positive
PushT	3074	blur	0.781	0.959	4.0	0.729	nonpositive / positive
PushT	3074	resize	1.173	0.939	-19.7	-0.577	nonpositive / nonpositive
Reacher	3072	blur	0.155	0.990	50.7	2.433	positive / positive
Reacher	3072	resize	0.210	0.990	29.7	2.433	positive / positive
Reacher	3073	blur	0.176	0.990	52.3	2.433	positive / positive
Reacher	3073	resize	0.207	0.990	40.0	2.433	positive / positive
Reacher	3074	blur	0.190	0.990	44.7	2.433	positive / positive
Reacher	3074	resize	0.195	0.990	33.3	2.433	positive / positive
Cube	3072	blur	1.447	0.330	-2.7	-1.488	nonpositive / nonpositive
Cube	3072	resize	1.166	0.530	1.0	-0.552	nonpositive / nonpositive
Cube	3073	blur	1.577	0.260	2.3	-1.923	nonpositive / nonpositive
Cube	3073	resize	1.218	0.560	-1.3	-0.728	nonpositive / nonpositive
Cube	3074	blur	1.220	0.660	-3.0	-0.735	nonpositive / nonpositive
Cube	3074	resize	1.040	0.770	-2.3	-0.134	nonpositive / nonpositive

F Local Gaussian Sensitivity as a Mechanistic Interpretation

For a small isotropic observation perturbation $\delta x \sim \mathcal{N}(0, \sigma^2 I)$, first-order linearization of encoder E followed by rollout map G gives

$$\mathbb{E}\big[\|\Delta G\|_2^2\big] \approx \sigma^2 \|J_G J_E\|_F^2. \tag{16}$$

This explains one mechanism by which Gaussian noise training can contract ACPC: it reduces the composed local sensitivity of the encoder–predictor path. It is a local approximation, not a global robustness bound and not a separate contribution.

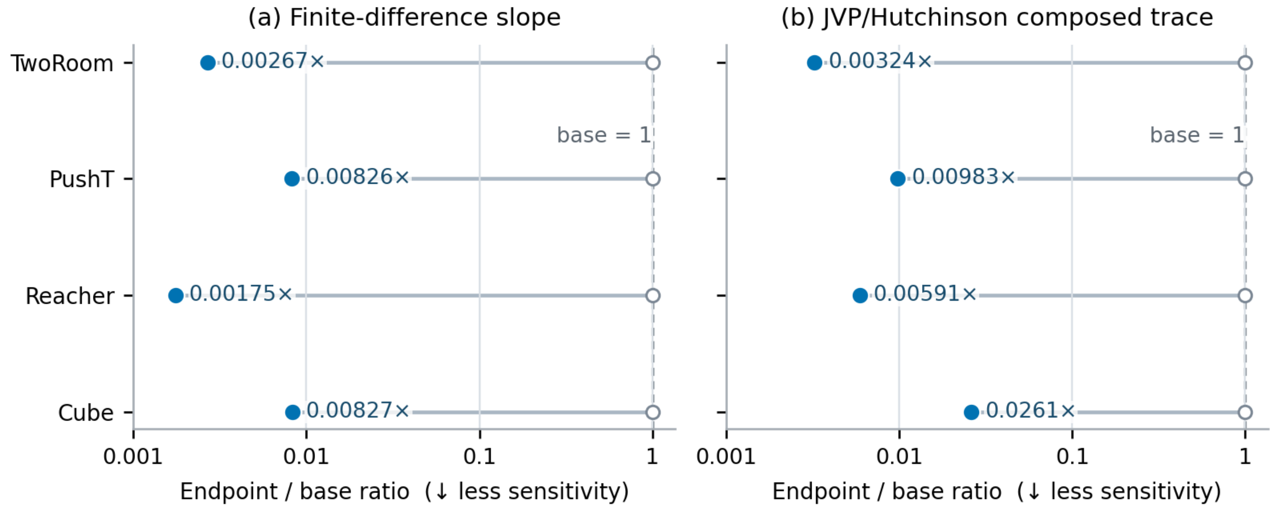

Figure 6: Endpoint/base local-sensitivity ratios from finite differences and a JVP/Hutchinson estimate of the composed Jacobian trace. Values below one indicate contraction after Gaussian noise training. The two estimators agree on the direction for every task; absolute ratios are mechanistic diagnostics, not closed-loop guarantees.

G Fixed-Pool Candidate Stability

The formal panel is behavior blind and uses seeds 3072/3073/3074 base/endpoint checkpoints for all four tasks. Fixed-pool shards capture the actual ordered step-zero proposal, recompute both branches through one shared cost path, and record exact same-winner and elite-set events for 100 histories per task, seed, and role at severities $0, 0.02, 0.05, 0.08$. Adaptive shards use a common zero proposal and common random numbers. The proposal-alignment induction is marked active only while all preceding elite updates remain certified.

For nominal final plan $\mathbf{a}$ and perturbed plan $\tilde{\mathbf{a}}$, the reported positive clean-history regret is

$$\big[C_h(\tilde{\mathbf{a}}) - C_h(\mathbf{a})\big]_+.$$

It measures decision degradation in model cost, not simulator return. First-action RMS is reported separately. The full-budget transfer keeps the same definitions but uses the same 16 fixed histories per task across roles and training seeds, $K = 300$, top-30 elites, and 30 CEM steps. Whole-shard runtimes and forward-call counts are retained for provenance, but without matched nominal-only timing they are not estimates of incremental diagnostic overhead.

Across 9,600 joined reduced-budget rows there are zero squared-distance-bound violations, zero false top-1 certificates, and zero false elite certificates. Identity probes have zero five-step displacement and zero first-action RMS; all identity updates remain aligned. These checks establish implementation soundness and are not additional independent statistical samples.

At severity 0.08, the sufficient top-1 certificate covers 10–70% of endpoint histories across task–seed blocks (and none of the fragile-base histories); elite-set coverage is 0–30% for endpoints. Coverage is lower than realized stability because the conditions certify rather than characterize decision invariance.

Table 9: Absolute planner outcomes at Gaussian severity 0.08, reported as mean ± sample SD across independent training runs after histories are averaged within run. Reduced-budget rows use 100 histories, $K = 64$, top-8 elites, and eight CEM steps; full-budget rows evaluate the same 16 fixed histories per task across roles and runs with $K = 300$, top-30 elites, and 30 steps. Regret is clean-history squared latent goal-cost, not closed-loop return, and its scale is task specific.

	Top-1 stability		Reduced regret		Full-budget regret	
Task	Base	Endpoint	Base	Endpoint	Base	Endpoint
TwoRoom	0.13±0.03	0.93±0.02	203.81±9.97	8.47±2.75	224.97±30.56	1.75±1.44
PushT	0.13±0.09	0.92±0.01	167.50±29.53	4.30±0.28	220.14±32.97	7.91±4.23
Reacher	0.07±0.01	0.96±0.04	281.12±20.24	0.47±0.14	269.44±22.21	0.28±0.20
Cube	0.02±0.02	0.92±0.03	255.72±12.28	1.31±0.11	265.03±28.08	0.63±0.22
Equal-task mean	0.09±0.03	0.93±0.01	227.04±16.92	3.64±0.68	244.89±27.08	2.65±1.43

H Qualitative Neighborhood Visualization

The two-dimensional view complements the quantitative analysis by showing how repeated perturbations of the same anchors are arranged before and after noise training. Because t-SNE does not preserve metric geometry, all reported evidence remains in the original projected-rollout space.

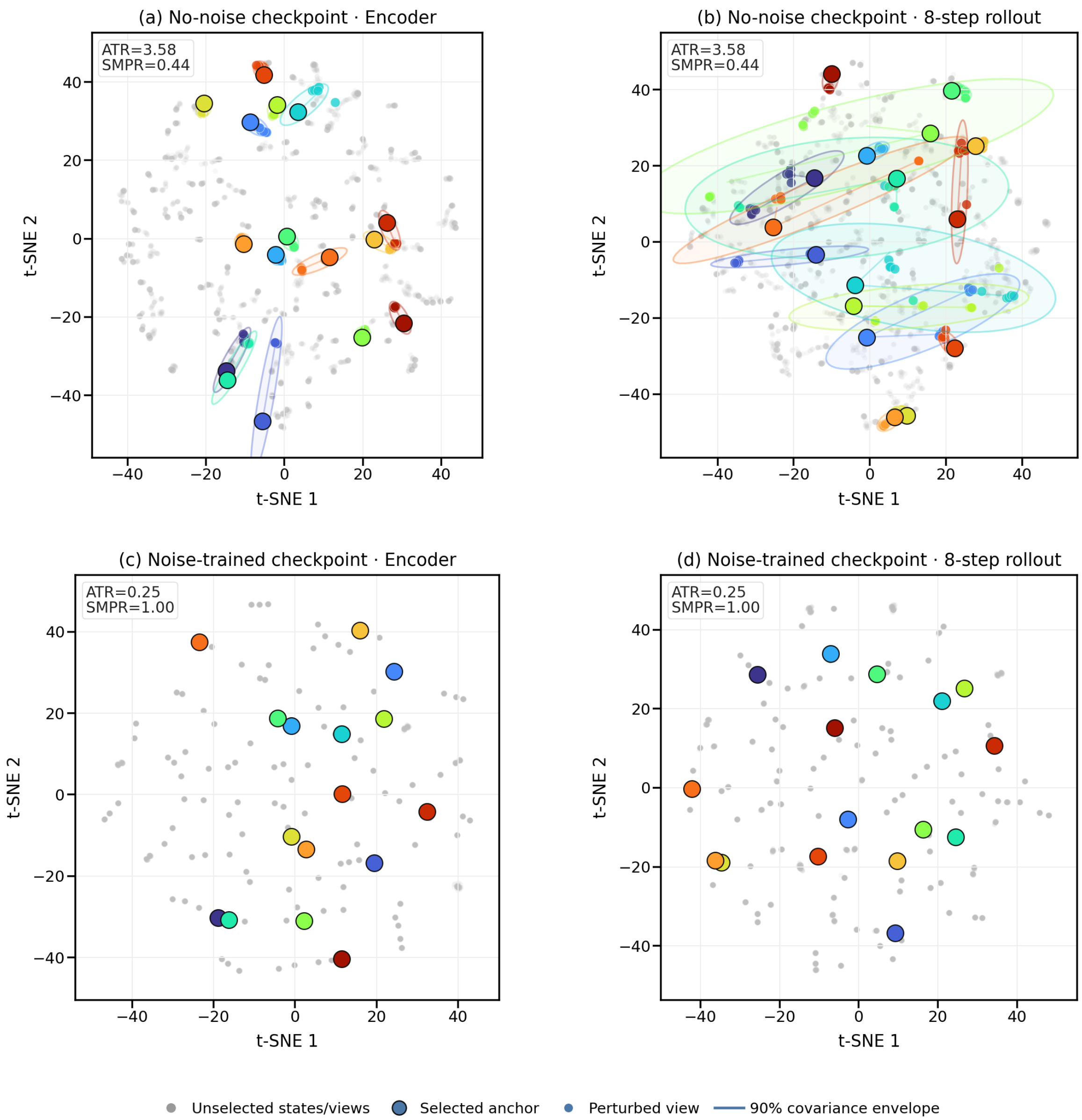

Figure 7: Qualitative PushT neighborhoods for matched perturbations at the no-noise and $\sigma_{\max} = 0.08$ checkpoints, in encoder and eight-step rollout spaces. t-SNE is not metric preserving: no displayed distance, area, or overlap enters a claim; ATR/SMPR are computed in the original space.